

Composition

Each capsule contains Fluoxetine hydrochloride 22.4mg corresponding to 20mg fluoxetine base

Product Description

Light green and buff, hard gelatin capsules, size 3, containing odourless powder.

Pharmacodynamics

Fluoxetine is a selective inhibitor of serotonin reuptake, and this probably accounts for the mechanism of action. Fluoxetine has practically no affinity to other receptors such as α 1-, α 2-, and β -adrenergic; serotonergic; dopaminergic;

histaminergic; muscarinic; and GABA receptors.

Pharmacokinetics

Absorption: Fluoxetine is well absorbed from the gastro-intestinal tract after oral administration.

The bioavailability is not affected by food intake.

Distribution: Fluoxetine is extensively bound to plasma proteins (about 95%) and it is widely distributed (volume of distribution: 20-40 l/kg). Steady-state plasma concentrations are achieved after dosing for several weeks. Steady-state concentrations after prolonged dosing are similar to concentrations seen at 4 to 5 weeks.

Metabolism: Fluoxetine has a non-linear pharmacokinetic profile with first pass liver effect. Maximum plasma concentration is generally achieved 6 to 8 hours after administration. Fluoxetine is extensively metabolised by the polymorphic enzyme CYP2D6. Fluoxetine is primarily metabolised by the liver to the active metabolite norfluoxetine (desmethylfluoxetine), by desmethylation.

Elimination: The elimination half-life of fluoxetine is 4 to 6 days and for norfluoxetine 4 to 16 days. These long half-lives are responsible for persistence of the drug for 5-6 weeks after discontinuation. Excretion is mainly (about 60%) via the kidney. Fluoxetine is secreted into breast milk.

At-risk populations

Elderly: Kinetic parameters are not altered in healthy elderly when compared to younger subjects.

Hepatic insufficiency: In case of hepatic insufficiency (alcoholic cirrhosis), fluoxetine and norfluoxetine half-lives are increased to 7 and 12 days, respectively. A lower or less frequent dose should be considered.

Renal insufficiency: After single-dose administration of fluoxetine in patients with mild, moderate, or complete (anuria) renal insufficiency, kinetic parameters have not been altered when compared to healthy volunteers. However, after repeated administration, an increase in steady-state plateau of plasma concentrations may be observed.

Indication

- Depression: Treatment of symptoms of depressive illness, with or without associated anxiety symptoms.
- Obsessive-compulsive disorder.
- Premenstrual dysphoric disorder (PMDD).
- Diagnosis of PMDD: The essential diagnostic features of PMDD are clear and established cyclicly (occurring during the last week of the luteal phase in most menstrual cycles) of symptoms e.g. depressed mood, anxiety, affective lability, accompanied by impairment in social and/or occupational function and physical symptoms (e.g. breast tenderness or swelling, headaches, joint or muscle pain, a sensation of bloating, weight gain), all of which must be severe. This syndrome should be distinguished from the commoner 'pre-menstrual tension' distinguished from PMDD by milder symptoms and less impact on normal activities' and from any coexisting psychiatric disorder.

Recommended Dose

For oral administration to adults only.

Depression, with or without Associated Anxiety Symptoms: *Adults and the Elderly:* Recommended Dose: 20 mg/day.

Obsessive-Compulsive Disorder: *Adults and the Elderly:* 20-60 mg/day. Recommended Initial Dose: 20 mg/day. Although there may be an increased potential for side effects at higher doses, a dose increase may be considered after several weeks if there is no response.

Premenstrual Dysphoric Disorder (PMDD):

Recommended Dose: 20 mg/day. Initial treatment should be limited to 6 months, after which patients should be re-assessed regarding the benefit of continued therapy.

All Indications: The recommended dose may be increased or decreased. Doses >80 mg/day have not been systematically evaluated.

Fluoxetine may be administered with or without food. When dosing is stopped, active drug substances will persist in the body for weeks. This should be borne in mind when starting or stopping treatment. Dosage tapering is unnecessary in most patients.

A lower or less frequent dose should be considered in patients with hepatic impairment, concurrent diseases, or who are taking multiple medications.

Use in children is not recommended as safety and efficacy have not been established.

Mode of Administration

Oral

Contraindication

Hypersensitivity to fluoxetine or to any of its excipients.

Monoamine oxidase inhibitors: Cases of serious and sometimes fatal reactions have been reported in patients receiving an SSRI in combination with a monoamine oxidase inhibitor (MAOI), and in patients who have recently discontinued an SSRI and have

been started on a MAOI. Treatment of fluoxetine should only be started 2 weeks after discontinuation of an irreversible MAOI.

Some cases presented with features resembling serotonin syndrome (which may resemble, and be diagnosed as, neuroleptic malignant syndrome). Cyproheptadine or dantrolene may benefit patients experiencing such reactions. Symptoms of a drug interaction with a MAOI include: hyperthermia, rigidity, myoclonus, autonomic instability with possible rapid fluctuations of vital signs, mental status changes that include confusion, irritability, and extreme agitation, progressing to delirium and coma.

Therefore, fluoxetine is contra-indicated in combination with a non-selective MAOI. Similarly, at least 5 weeks should elapse after discontinuing fluoxetine treatment before starting a MAOI. If fluoxetine has been prescribed chronically and/or at a high dose, a longer interval should be considered.

The combination is not recommended. Treatment with fluoxetine can be initiated the following day after discontinuation of a reversible MAOI (eg, moclobemide).

Warnings and Precautions

Warnings

Suicidality in Children and Adolescents

- Antidepressants increase the risk of suicidal thinking and behavior (suicidality) in children and adolescents with major depressive disorder (MDD) and other psychiatric disorders.
- Anyone considering the use of an antidepressant in a child or adolescent for any clinical use must balance the risk of increased suicidality with the clinical need.
- Patients who are started on therapy should be observed closely for clinical worsening, suicidality, or unusual changes in behavior.
- Families and caregivers should be advised to closely observe the patient and to communicate with the prescriber.

Rash and allergic reactions: Rash, anaphylactoid events, and progressive systemic events, sometimes serious (involving skin, kidney, liver, or lung), have been reported. Upon the appearance of rash or of other allergic phenomena for which an alternative aetiology cannot be identified, fluoxetine should be discontinued.

Precautions

Seizures: Seizures are a potential risk with antidepressant drugs. Therefore, as with other antidepressants, fluoxetine should be introduced cautiously in patients who have a history of seizures. Treatment should be discontinued in any patient who develops seizures or where there is an increase in seizure frequency. Fluoxetine should be avoided in patients with unstable seizure disorders/epilepsy, and patients with controlled epilepsy should be carefully monitored.

Mania: Antidepressants should be used with caution in patients with a history of mania/hypomania. As with all antidepressants, fluoxetine should be discontinued in any patient entering a manic phase.

Hepatic/renal function: Fluoxetine is extensively metabolised by the liver and excreted by the kidneys. A lower dose, eg, alternate day dosing, is recommended in patients with significant hepatic dysfunction. When given fluoxetine 20mg/day for 2 months, patients with severe renal failure (GFR <10ml/min) requiring dialysis showed no difference in plasma levels of fluoxetine or norfluoxetine compared to controls with normal renal function.

Cardiac disease: No conduction abnormalities that resulted in heart block were observed in the ECG of 312 patients who received fluoxetine in double-blind clinical trials. However, clinical experience in acute cardiac disease is limited, therefore caution is advisable.

Weight loss: Weight loss may occur in patients taking fluoxetine but it is usually proportional to baseline body weight.

Diabetes: In patients with diabetes, treatment with an SSRI may alter glycaemic control. Hypoglycaemia has occurred during therapy with fluoxetine, and hyperglycaemia has developed following discontinuation. Insulin and/or oral hypoglycaemic dosage may need to be adjusted.

Suicide: As improvement may not occur during the first few weeks of treatment, in common with all antidepressants, patients should be closely monitored during this period. The possibility of a suicide attempt is inherent in depression and may persist until significant remission occurs. It is general clinical experience with all therapies for depression that the risk of suicide may increase in the early stages of recovery.

Haemorrhage: There have been reports of cutaneous bleeding abnormalities, such as ecchymosis and purpura, with SSRIs. Ecchymosis has been reported as an infrequent event during treatment with fluoxetine. Other haemorrhagic manifestations (eg, gynaecological haemorrhages, gastro-intestinal bleedings, and other cutaneous or mucous bleedings) have been reported rarely. Caution is advised in patients taking SSRIs, particularly in concomitant use with oral anticoagulants, drugs known to affect platelet function (eg, atypical antipsychotics, such as clozapine, phenothiazines, most TCAs, aspirin, NSAIDs), or other drugs that may increase risk of bleeding, as well as in patients with a history of bleeding disorders.

Electroconvulsive therapy (ECT): There have been rare reports of prolonged seizures in patients on fluoxetine receiving ECT treatment, therefore caution is advisable.

St John's Wort: An increase in serotonergic effects, such as serotonin syndrome, may occur when selective serotonin reuptake inhibitors and herbal preparations containing St John's Wort (*Hypericum perforatum*) are used together.

On rare occasions, development of a serotonin syndrome or neuroleptic malignant syndrome-like events have been reported in association with treatment of fluoxetine, particularly when given in combination with other serotonergic (among others, L-tryptophan) and/or neuroleptic drugs. As these syndromes may result in potentially life-threatening conditions, treatment with fluoxetine should be discontinued if such events (characterised by clusters of symptoms, such as hyperthermia, rigidity, myoclonus, autonomic instability with possible rapid fluctuations of vital signs, mental status changes, including confusion, irritability, extreme agitation, progressing to delirium and coma) occur, and supportive symptomatic treatment should be initiated.

Sexual dysfunction: Selective serotonin reuptake inhibitors (SSRIs)/serotonin norepinephrine reuptake inhibitors (SNRIs) may cause symptoms of sexual dysfunction. There have been reports of long-lasting sexual dysfunction where the symptoms have continued despite discontinuation of SSRIs/SNRIs.

Effects on ability to drive and use machines

Although fluoxetine has been shown not to affect psychomotor performance in healthy volunteers, any psychoactive drug may impair judgement or skills. Patients should be advised to avoid driving a car or operating hazardous machinery until they are reasonably certain that their performance is not affected.

Interactions

Half-life: The long elimination half-lives of both fluoxetine and norfluoxetine should be borne in mind when considering pharmacodynamic or pharmacokinetic drug interactions.

Monoamine oxidase inhibitors: See Contraindications

Not recommended combinations: MAOI-A (See Contraindications)

Combinations requiring precautions for use: MAOI-B (selegiline); risk of serotonin syndrome. Clinical monitoring is recommended.

Phenytoin: Changes in blood levels have been observed when combined with fluoxetine. In some cases manifestations of toxicity have occurred. Consideration should be given to using conservative titration schedules of the concomitant drug and to monitoring clinical status.

Serotonergic drugs: Co-administration with serotonergic drugs (eg, tramadol, triptans) may increase the risk of serotonin syndrome. Use with triptans carries the additional risk of coronary vasoconstriction and hypertension.

Lithium and tryptophan: There have been reports of serotonin syndrome when SSRIs have been given with lithium or tryptophan and, therefore, the concomitant use of fluoxetine with these drugs should be undertaken with caution. When fluoxetine is used in combination with lithium, closer and more frequent clinical monitoring is required.

CYP2D6 isoenzyme: Because fluoxetine's metabolism (like tricyclic antidepressants and other selective serotonin antidepressants) involves the hepatic cytochrome CYP2D6 isoenzyme system, concomitant therapy with drugs also metabolised by this enzyme system may lead to drug interactions. Concomitant therapy with drugs predominantly metabolised by this isoenzyme, and which have a narrow therapeutic index (such as flecainide, encainide, carbamazepine, and tricyclic antidepressants), should be initiated at or adjusted to the low end of their dose range. This will also apply if fluoxetine has been taken in the previous 5 weeks.

Oral anticoagulants: Altered anticoagulant effects (laboratory values and/or clinical signs and symptoms), with no consistent pattern, but including increased bleeding, have been reported uncommonly when fluoxetine is co-administered with oral anticoagulants. Patients receiving warfarin therapy should receive careful coagulation monitoring when fluoxetine is initiated or stopped

Electroconvulsive therapy (ECT): There have been rare reports of prolonged seizures in patients on fluoxetine receiving ECT treatment, therefore caution is advisable.

Alcohol: In formal testing, fluoxetine did not raise blood alcohol levels or enhance the effects of alcohol. However, the combination of SSRI treatment and alcohol is not advisable.

St John's Wort: In common with other SSRIs, pharmacodynamic interactions between fluoxetine and the herbal remedy St John's Wort (*Hypericum perforatum*) may occur, which may result in an increase of undesirable effects.

Pregnancy and Lactation

Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress, and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalisation.

Fluovex should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

Pregnancy: Data on a large number of exposed pregnancies do not indicate a teratogenic effect of fluoxetine. Fluoxetine can be used during pregnancy, but caution should be exercised, especially during late pregnancy or just prior to the onset of labour, since the following effects have been reported in neonates: irritability, tremor, hypotonia, persistent crying, difficulty in sucking or in sleeping. These symptoms

may indicate either serotonergic effects or a withdrawal syndrome. The time to occur and the duration of these symptoms may be related to the long half-life of fluoxetine (4-6 days) and its active metabolite, norfluoxetine (4-16 days).

Lactation: Fluoxetine and its metabolite, norfluoxetine, are known to be excreted in human breast milk. Adverse events have been reported in breast-feeding infants. If treatment with fluoxetine is considered necessary, discontinuation of breast-feeding should be considered; however, if breast-feeding is continued, the lowest effective dose of fluoxetine should be prescribed.

Observational data indicate an increased risk (less than 2-fold) of postpartum haemorrhage following SSRI/SNRI exposure within the month prior to birth.

Side Effects

Undesirable effects may decrease in intensity and frequency with continued treatment and do not generally lead to cessation of therapy.

In common with other SSRIs, the following undesirable effects have been seen:

Body as a whole: Hypersensitivity (eg, pruritus, rash, urticaria, anaphylactoid reaction, vasculitis, serum sickness-like reaction, angioedema), chills, serotonin syndrome, photosensitivity, and, very rarely, toxic epidermal necrolysis (Lyell syndrome).

Digestive system: Gastro-intestinal disorders (eg, diarrhoea, nausea, vomiting, dyspepsia, dysphagia, taste perversion), dry mouth. Abnormal liver function tests have been reported rarely. Very rare cases of idiosyncratic hepatitis.

Nervous system: Headache, sleep abnormalities (eg, abnormal dreams, insomnia), dizziness, anorexia, fatigue (eg, somnolence, drowsiness), euphoria, transient abnormal movement (eg, twitching, ataxia, tremor, myoclonus), seizures, and psychomotor restlessness. Hallucinations, manic reaction, confusion, agitation, anxiety and associated symptoms (eg, nervousness), impaired concentration and thought process (eg, depersonalisation), panic attacks (these symptoms may be due to the underlying disease), and, very rarely serotonin syndrome.

Urogenital system: Urinary retention, urinary frequency.

Reproductive disorders: Sexual dysfunction (delayed or absent ejaculation, anorgasmia), priapism, galactorrhoea.

Miscellaneous: Alopecia, yawn, abnormal vision (eg, blurred vision, mydriasis), sweating, vasodilatation, arthralgia, myalgia, postural hypotension, ecchymosis. Other haemorrhagic manifestations (eg, gynaecological haemorrhages, gastro-intestinal bleedings, and other cutaneous or mucous bleedings) have been reported rarely.

Hyponatraemia: Hyponatraemia (including serum sodium below 110mmol/l) has been rarely reported and appeared to be reversible when fluoxetine was discontinued. Some cases were possibly due to the syndrome of inappropriate antidiuretic hormone secretion. The majority of reports were associated with older patients, and patients taking diuretics or otherwise volume depleted.

Respiratory system: Pharyngitis, dyspnoea. Pulmonary events (including inflammatory processes of varying histopathology and/or fibrosis) have been reported rarely. Dyspnoea may be the only preceding symptom.

When stopping treatment, withdrawal symptoms have been reported in association with SSRIs, although the available evidence does not suggest this is due to dependence. Common symptoms include dizziness, paraesthesia, headache, anxiety, and nausea, the majority of which are mild and self-limiting. Fluoxetine has been only rarely associated with such symptoms. Plasma fluoxetine and norfluoxetine concentrations decrease gradually at the conclusion of therapy, which makes dosage tapering unnecessary in most patients.

Symptoms And Treatment of Overdose

Cases of overdose of fluoxetine alone usually have a mild course. Symptoms of overdose have included nausea, vomiting, seizures, cardiovascular dysfunction ranging from asymptomatic arrhythmias to cardiac arrest, pulmonary dysfunction, and signs of altered CNS status ranging from excitation to coma. Fatality attributed to overdose of fluoxetine alone has been extremely rare. Cardiac and vital signs monitoring are recommended, along with general symptomatic and supportive measures. No specific antidote is known.

Forced diuresis, dialysis, haemoperfusion, and exchange transfusion are unlikely to be of benefit. Activated charcoal, which may be used with sorbitol, may be as or more effective than emesis or lavage. In managing overdosage, consider the possibility of multiple drug involvement. An extended time for close medical observation may be needed in patients who have taken excessive quantities of a tricyclic antidepressant if they are also taking, or have recently taken, fluoxetine.

Storage Condition

Store below 30°C protected from light and humidity.

Pack Size: 30 capsules

Manufacturer

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